

DAT246 Exam Summary Sheet

(Bayesian Data Analysis & Empirical Software Engineering)

1■■■ Bayesian Data Analysis Workflow

Steps: DAG → Model → Priors → Prior Check → Fit (MCMC) → Diagnostics → Posterior Check → Compare (WAIC/LOO) → Interpret & Iterate.

Keywords: iterative, causal reasoning, prior/posterior predictive checks.

2■■■ Avoiding Overfitting

Regularizing priors: Shrink parameters.

Information criteria: WAIC, PSIS-LOO measure predictive accuracy.

Multilevel models: Partial pooling shares info between groups.

3■■■ Validity & Research Design

Experiments: High internal validity, low external.

Observations: High external, low internal.

Tension: Control ↔ Realism.

4■■■ Information Theory & Model Comparison

Entropy: $H(p) = -\sum p \log p$ (uncertainty).

KL divergence: $D_{KL}(p||q) = \sum p \log(p/q)$ (information loss).

WAIC/LOO: Approximate $-KL$ for predictive accuracy.

5■■■ Generalized Linear Models (GLMs) + Max Entropy

Max Ent Principle: Most conservative distribution consistent with constraints.

Examples: Normal (mean,var), Binomial (binary), Poisson (count).

Form: $g(\mu) = \alpha + \beta x \rightarrow$ connects linear predictor & likelihood.

6■■■ MCMC & Diagnostics

Goal: Approximate posterior $p(\theta|y)$.

Metropolis: Propose–Accept–Repeat.

HMC: Uses gradients.

Diagnostics: $R^2 \approx 1$, $n_{eff} \uparrow$, trace plots mix, no divergents.

7■■■ Multilevel Models & Pooling

Complete: All groups same.

No: Independent fits.

Partial: Shared hyperpriors $\rightarrow$ shrinkage.

Form: $\alpha_j \sim N(\alpha, \sigma)$; $\alpha \sim N(0,1)$; $\sigma \sim \text{Exp}(1)$.

8■■■ Mixture & Zero-Inflated Models

Mixture: Combine components $\rightarrow$ over-dispersion (e.g. Beta-Binomial, Neg-Binomial).

Zero-inflated: Add extra zero process (ZIP).

Difference: Mixtures handle variance; ZI handles zeros.

9■■■ Covariance & Gaussian Processes

MVN: $[\alpha_j \beta_j]^T \sim \text{MVN}([\alpha \beta], \Sigma)$.

Σ : var-covar matrix with ρ correlation.

LKJ(η): Prior for correlations ($\eta > 1 \rightarrow$ weak corr).

GP: $f(x) \sim \text{GP}(m(x), k(x, x')) \rightarrow$ continuous covariance.